NEX

Nexchain:

A Risk-Aware Smart Contract Network

Technical Whitepaper

JULY 2026

Notice

Notice. This document is a technical description of the Nexchain network and its design. It is informational and does not constitute financial advice, an offer of securities, or a solicitation to invest. It is not a crypto-asset white paper within the meaning of Regulation (EU) 2023/1114 (MiCA) and should not be relied upon as one. Forward-looking statements describe design intent and planned work; they are not guarantees. Quantitative performance characteristics, where mentioned, are design targets rather than measured production results and are labeled as such. See Risk Factors.

Abstract

Nexchain is an EVM-compatible smart contract network whose defining property is native transaction risk intelligence: before a user commits a transaction, the network's tooling evaluates it and returns a clear risk assessment, surfaced ahead of signing. Where existing networks treat user safety as an optional, third-party add-on, Nexchain integrates risk awareness into the core network experience as a first-class service. This paper describes the problem, the network's layered architecture, the risk intelligence pipeline that is its central contribution, the token that coordinates the system, the current security and decentralization posture, and the material risks. The paper is deliberate about distinguishing what is implemented from what is designed or planned. ---

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1. Introduction

Blockchain networks have advanced substantially in throughput, tooling, and capital formation. The experience of *using* one, however, remains hostile to ordinary participants. A transaction presented for signature is, in practice, an opaque payload: approving a malicious contract, interacting with a wallet drainer, or signing an exploit-laden call looks - at the decisive moment - identical to a safe action. The resulting losses are measured in billions of dollars per year across the industry.

Nexchain begins from a single thesis: **risk assessment should be a native property of the network experience, not an afterthought bolted on by external extensions.** By scoring a transaction for risk and presenting that score, with its contributing factors, before the user signs, the network converts a moment of blind trust into an informed decision. The user remains in control; the network's contribution is context they would not otherwise have.

This paper describes how Nexchain is built to deliver that property, and is explicit throughout about the maturity of each component.

2. Background & Motivation

2.1 The safety gap

Two mechanisms exist today to protect users, and both are incomplete. Contract audits address security at deploy time but say nothing about a specific transaction a user is about to sign months later. Wallet extensions and simulation services attempt sign-time safety, but they are optional, fragmented across vendors, inconsistent in coverage, and operate without network-level context. The user who most needs protection - the one who does not install specialist tooling - receives none.

2.2 The case for a native approach

Integrating risk awareness at the network and tooling layer makes the signal consistent (the same evaluation logic applies to every participant), always present (it is not an optional install), and improvable over time (it learns from on-chain patterns the whole network observes). This is a structural difference from networks where safety tooling is an opt-in accessory.

2.3 Design principles

- **Inform, don't gate.** The network surfaces risk; it does not unilaterally block users. Agency stays with the user.
- **Explainability over verdicts.** A score a user can interrogate is more useful than an opaque label. The system aims to expose *why*.
- **Separation of concerns.** Risk evaluation runs adjacent to, not inside, the consensus-critical path, so it can evolve quickly and fail gracefully.
- **Honest disclosure.** The network's maturity - what is audited, what is decentralized, what is live - is stated plainly. Overstating verified properties is itself a risk.

3. Network Model

This section describes the network abstractly, in the manner of a state-transition system, to fix terminology used later.

3.1 State

Nexchain maintains a global **state** consistent with EVM conventions: an account-based mapping in which each account carries a balance, a nonce, and, for contract accounts, associated code and storage. The aggregate of all account states constitutes the network state at a given height.

3.2 State transitions

A **transaction** is a signed instruction that, when applied, deterministically transforms the state from one valid configuration to another. Execution is deterministic: given the same pre-state and the same transaction, every honest node computes the same post-state. Invalid transactions (insufficient balance, malformed calls, gas exhaustion) are rejected by the transition rules rather than producing an undefined state.

3.3 Risk annotation

Distinct from the state transition itself, Nexchain associates with a candidate transaction a **risk annotation** - a structured assessment produced by the risk intelligence layer (Section 5). The annotation is advisory metadata consumed before signing; it is not part of the consensus-critical state transition and does not alter execution semantics. This separation is intentional and is revisited in Section 4.

3.4 Participants

The network model distinguishes ordinary **users** (who submit transactions), **developers** (who deploy and operate contracts and applications), and **operators** (who run the infrastructure that orders, executes, and serves the network). The current distribution of operator responsibility is described candidly in Section 10.

4. System Architecture

Following the principle of separation of concerns, Nexchain is organized into four conceptual layers. Each can evolve and scale with minimal coupling to the others.

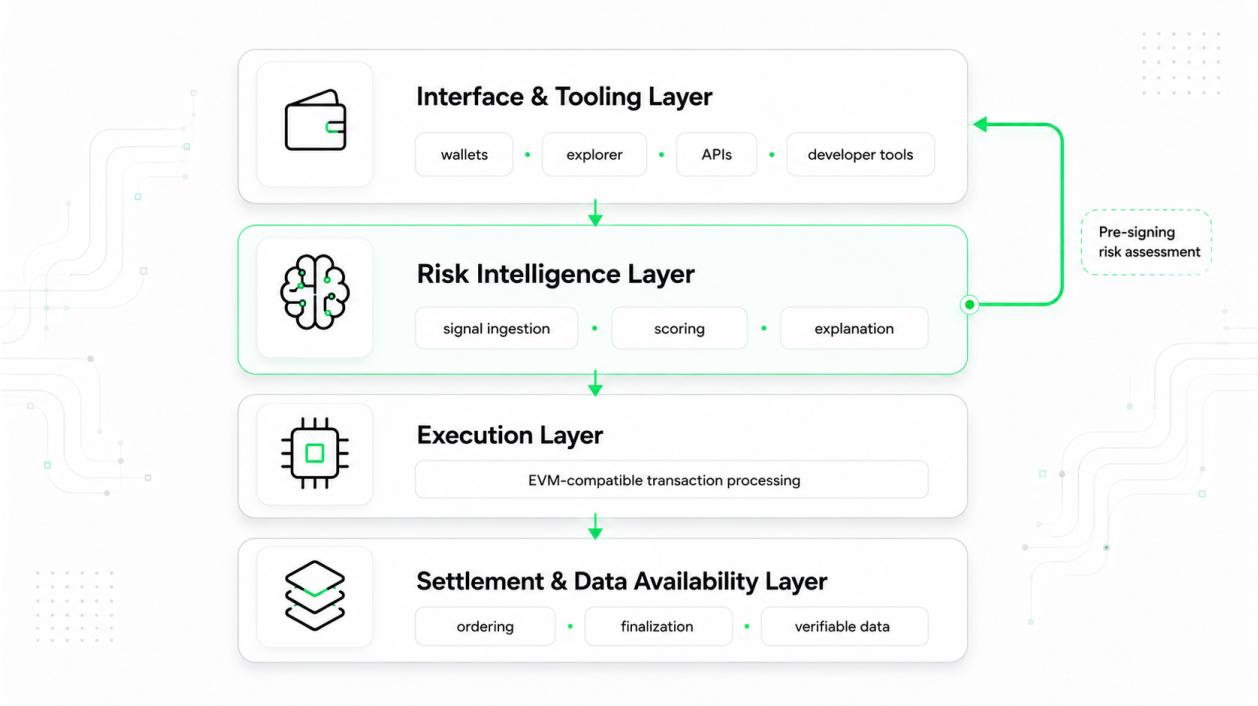


Figure 1. Nexchain's four-layer system architecture.

4.1 Settlement & data availability layer

Transactions are ordered and finalized through the network's settlement process, after which the associated data is published in a form that can be independently verified. The design goal is reconstructability: any party should be able to rebuild and check network state from published data rather than trusting an operator's assertion. Data availability is treated as a first-class concern precisely because the network's risk and transparency guarantees depend on it - a risk annotation is only as trustworthy as the ability to inspect the data behind it.

4.2 Execution layer

The execution layer is EVM-compatible. Contracts written in standard Solidity, existing development frameworks, and common wallets operate with little or no modification. This is a deliberate, pragmatic choice: it lowers the migration barrier for developers and lets Nexchain inherit a mature, battle-tested tooling base rather than imposing a new paradigm.

4.3 Risk intelligence layer

The risk intelligence layer evaluates candidate transactions and produces the risk annotations defined in Section 3.3. Critically, it operates **adjacent to** the consensus-critical execution path, not inside it. Two consequences follow: the scoring models can be retrained, extended, and improved without changing core network behavior; and a degraded or unavailable scoring service does not halt the network - it degrades gracefully to an unscored state rather than blocking transactions. This layer is detailed in Section 5.

4.4 Interface & tooling layer

Wallets, the block explorer, programmatic APIs, and developer tooling expose the network and - importantly - its risk signals to users and builders. Risk intelligence that could not be surfaced cleanly to a user at the moment of signing would be of little value; this layer is where the network's core property becomes tangible.

5. Risk Intelligence Pipeline

This is the network's central technical contribution and the analogue, in Nexchain's design, of the consensus protocol in a conventional L1 paper: the component that most defines the system.

5.1 Overview

Conceptually, risk evaluation is a four-stage pipeline operating on a candidate transaction before signature:

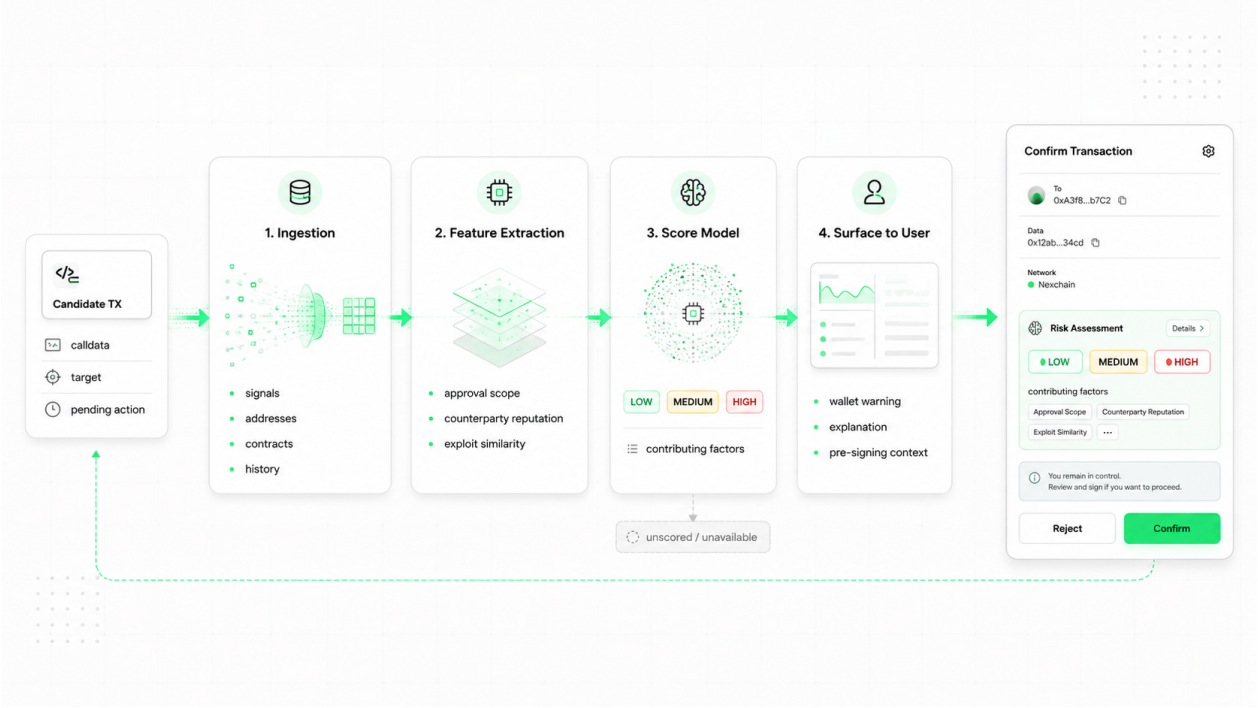


Figure 2. Candidate transaction risk intelligence pipeline.

5.2 Signal ingestion

The pipeline gathers inputs relevant to risk: the transaction's calldata and target, properties of the addresses and contracts involved, historical interaction patterns, and known indicators of malicious activity (for example, association with flagged addresses or resemblance to documented exploit signatures). The set of signal sources is designed to expand over time; new sources can be added without changes to downstream consensus or execution.

5.3 Feature extraction

Raw signals are transformed into structured features the model can reason over - for example, the scope of a token approval, the reputation of a counterparty, or the similarity of a call to known exploit patterns. Feature extraction is where domain knowledge about on-chain attack patterns is encoded.

5.4 Scoring

A model evaluates the extracted features and produces a risk assessment expressed in clear terms - low / medium / high - together with the principal contributing factors. The scoring component is designed to be versioned and replaceable: improvements to the model do not require changes elsewhere in the system.

5.5 Surfacing & explainability

The assessment is delivered to the user through the wallet or application interface ahead of signing, accompanied by the factors driving it. Explainability is a first-class requirement: the system aims to answer *why* a transaction was scored as it was, not merely to emit a label. A signal the user can interrogate is one they can act on responsibly.

5.6 Failure behavior

Because the pipeline sits outside the consensus-critical path, its unavailability is a degradation, not an outage. If scoring cannot be produced, transactions proceed in an unscored state and the interface communicates that the assessment is unavailable - the network does not halt, and the user is not silently led to believe an unscored transaction was scored safe.

Status: the risk intelligence pipeline is in active development and integration. Signal sources and model coverage are expected to expand over time. Descriptions in this section reflect design intent.

6. Execution & State

6.1 Account model

State is maintained as an account-based model consistent with EVM conventions (Section 3.1). Externally-owned accounts and contract accounts share the same address space; contract accounts additionally carry code and persistent storage.

6.2 Determinism

Execution is deterministic and gas-metered. Gas bounds computation, preventing unbounded execution and pricing the cost of state changes. Standard EVM semantics govern how transactions consume gas and mutate state.

6.3 Compatibility

By adhering to EVM semantics and standard interfaces, Nexchain remains compatible with the existing ecosystem of Solidity contracts, token standards, wallets, and development tooling. This compatibility is a design constraint, not an incidental property: it is what allows existing applications and developer knowledge to transfer.

7. Developer Platform

> Concrete endpoints, SDK packages, contract addresses, and parameters will be published in dedicated developer documentation as they are finalized. This section describes intended design.

Because the execution environment is EVM-compatible, the developer workflow is intended to be familiar: contracts in standard Solidity, deployment via common frameworks, and connection through standard RPC interfaces and wallets.

7.3 Developer and integration surface

What distinguishes building on Nexchain is programmatic access to risk intelligence.
The intended model exposes a risk interface for applications and developer tooling.

A conceptual integration flow:

1. Connect to the network via its RPC endpoint.
2. Build and submit transactions using standard EVM tooling.
3. Optionally request a risk assessment before prompting the user to sign.
4. Surface the returned assessment and its contributing factors in the application interface.

Supporting components include a public block explorer, node/RPC interfaces, a faucet and test environment.
SDKs, the risk API, and full integration documentation are planned ahead of mainnet.

8. Applications

Finance & DeFi. Native risk scoring can provide users with a pre-commit signal for contract interactions.

Consumer wallet safety. Applications can surface a warning before a user signs a risky transaction.

Compliance & monitoring. Risk signals can support auditable monitoring use cases.

Decentralized AI services. The network is designed to support applications combining on-chain logic with AI.

Governance. Token-based governance can support decisions over network parameters and treasury use.

9. Token (NEX)

> This section describes current token parameters. It does not predict or commit to future token value.
> The current allocation and unlock schedule are provided on the next page.

9.1 Overview

Token parameters and the current allocation table are shown on page 10.

9. Token (NEX)

This section describes the current token parameters. It does not predict or commit to future token value.

9.1 Overview

Parameter	Value
Token	NEX
Total Supply	1,000,000,000 NEX
Standard	EVM-compatible token
Public Sale unlock schedule	Published 5 October

Launch and market metrics are not published in this document.

9.2 Utility

NEX is the network's native asset, used for transaction fees, access to AI risk-intelligence features, staking, and governance participation.

9.3 Distribution

Allocation	Supply share	Tokens
Public Sale	30%	300,000,000
Treasury	15%	150,000,000
Ecosystem	15%	150,000,000
Team	10%	100,000,000
Liquidity	10%	100,000,000
Private	8%	80,000,000
Rewards / Burn	7%	70,000,000
Seed / Marketing	5%	50,000,000

9.4 Unlock schedule

Allocation	At TGE	Unlock schedule
Public Sale	-	Published 5 October
Treasury	0%	Linear over 36 months
Ecosystem	0%	Linear over 36 months
Team	0%	12-month cliff, then 24 months
Liquidity	-	Locked in liquidity pool
Private	0%	6-month cliff, then 18 months
Rewards / Burn	-	Rewards emissions
Seed / Marketing	0%	Vesting

Allocation size is not the amount unlocked. Unlocking follows a separate schedule.
The full Public Sale unlock schedule and final supply figures will be published on 5 October.

9.5 Buyback & burn

A portion of network fees may fund a buyback-and-burn program, subject to published protocol governance and implementation details. Any supply reduction will be reported through official updates.

9.6 Presale structure

Presale participation follows the current entry windows published on the official Nexchain website. Prices, launch details, and eligibility conditions are subject to official project updates.

Presale participation carries the same risks as any early-stage token purchase. Token value may fall, including to zero. Participants should conduct their own research before participating.

10. Security & Decentralization

Security is treated as a product feature, which includes being transparent about its current scope.

Audited. The NEX token contract has been audited by independent third-party security firms. This audit scope covers the token contract specifically.

Not yet audited. The broader protocol and network-level components have not yet undergone a full independent audit. Nexchain does not represent the full protocol as independently audited.

Decentralization maturity. Nexchain is at an early stage of decentralization. The network is currently developed and operated by the core team, with progressive decentralization planned over time.

11. Risk Factors

Prospective participants should consider the following non-exhaustive risks.

? **Early-stage technology.** Core components, including the risk intelligence pipeline and mainnet, are under development. Planned features may change, be delayed, or not ship as described.

? **Limited audit scope.** Only the token contract has been independently audited. Protocol-level components have not yet been audited and may contain undiscovered vulnerabilities.

? **Centralization.** The network is presently operated by the core team. Progressive decentralization is a goal, not a present state.

? **Risk-scoring limitations.** Risk annotations are advisory and probabilistic. They may produce false negatives or false positives. They do not guarantee safety and are not a substitute for user diligence.

? **Token price.** Token value is set by the market and may fall, including to zero. Launch price information is not a promise of value or return. No statement in this paper should be read as a price prediction.

? **Liquidity & market.** Early-stage tokens may have limited liquidity and high volatility.

? **Regulatory.** The regulatory treatment of crypto-assets is evolving across jurisdictions and may affect the network, the token, or participants' ability to access either.

? **Execution.** Roadmap dates and scope are targets and may change as development progresses.

This list does not capture every risk. Participants should conduct their own research and consider their own circumstances.

12. Roadmap

A phased journey from concept to a live network - built in the open, shipped in stages. Dates and scope are targets and may change.

Phase	Focus	Status
Q1 2025	Idea & Foundation - vision, team, design principles, initial paper	Completed
Q2 2025	Early Access & Community - token infrastructure, third-party audit, early access, channels	Completed
Q3 2025	Beta Testnet - first public testnet, explorer, faucet, early feedback	Completed
Q4 2025	Advanced Testnet - risk-awareness prototype, wallet & transaction flow testing, expanded docs	Completed
Q1 2026	Infrastructure Scaling - production-scale infrastructure, hardened tooling, groundwork for AI rollout	Completed
Q2 2026	Mainnet Preparation - mainnet candidate to staging, additional audits, developer tooling, partner onboarding	In Progress
Q3 2026	Token Launch - network goes live, Token Generation Event, staking & governance, builder programs	Upcoming

13. Conclusion

Nexchain is built around a problem every blockchain user faces: signing transactions without understanding their risk. By integrating risk intelligence into the core network experience - as a first-class, explainable, gracefully-degrading service rather than an optional add-on - Nexchain aims to make informed signing the default.

The work ahead is substantial: completing the risk pipeline integration, extending security audits to the full protocol, and launching mainnet. This paper will be revised as those milestones are reached, and should be read as a description of a system in active construction, with its maturity stated honestly throughout.

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